MIL-C-38226B (USAF)
9 March 1987
SUPERSEDING
MIL-C-38226A (USAF)
6 December 1968

MILITARY SPECIFICATION

CONTAINERS, POLYURETHANE, RIGID OR FLEXIBLE FOR PACKAGING SMALL ENGINES

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general requirements for polyurethane shipping container for packaging small engines not exceeding 500 pounds.

1.2 Classification. The polyurethane shipping containers shall be furnished in the following classes:

Class 1 - Level "A" pack, world-wide shipment and storage.

Class 2 - Level "C" pack, immediate use at the first domestic receiving activity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Air Force Packaging Evaluation Agency (HQ AFLC/DSTZ), Wright-Patterson AFB OH 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8115

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-B-22191	Barrier Materials, Transparent, Flexible, Heat Sealable
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-I-26860	Indicator, Humidity, Plug, Color Change
MIL-F-83671	Foam-In-Place Packaging Materials, General Specification for
MIL-V-27166	Valve, Pressure Equalizing, Gaseous Products

STANDARDS

Federal

FED-STD-101	Test Procedures for Packaging Materials

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-648	Design Criteria for Specialized Shipping Containers
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed on the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (ASTM).

ASTM E380 Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified by the contract or purchase order, a sample shall be subject to first article inspection (see 4.4 and 6.2).

3.2 Materials. The polyurethane may be rigid or flexible pre-foamed, foamed-in-place, or a combination of both. The materials shall conform to the applicable requirements of MIL-P-26514 or MIL- F-83671.

3.2.1 Uniformity. There shall be no openings larger than 1 inch in diameter and all such openings shall be separated by a firm material of 1 inch minimum thickness. Accumulations of unexpanded resin shall not be allowed.

3.2.2 Thickness. The foam between the engine preserved according to paragraph 3.3.1 and each face of the exterior container (top, bottom, sides and ends) shall be the minimum required to protect the engine. At no time shall the foams be permitted to make direct contact with the engine.

3.2.3 Density. The density of the foam shall not be less than 1.3 nor more than 2.5 pounds per cubic foot, depending on the characteristics of the engine to be protected (see 4.6.3).

3.3 Preliminary design, drawings and proposal. The preliminary design, drawing, and proposal, to utilize a polyurethane shipping container, shall be prepared and submitted to the procuring activity for approval.

3.3.1 Preservation. Engines shall be internally preserved in accordance with the applicable engine specifications. Externally, the engines shall be preserved as specified in 5.1.

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:07Z
Check the source to verify that this is the current version before use.

3.4 Design and construction. Class 2 containers shall be designed to provide protection for engines against shock, vibration and deterioration encountered during shipment to the first receiving activity for immediate use. Class 1 containers shall be designed to provide protection for engines against shock, vibration, and deterioration encountered during shipment and storage. The following sequence shall be used to test Class 1 and 2 containers (see paragraph 4.4 and 4.6) except that cycle exposure test is not required for Class 2.

a. Cornerwise Drop.

b. Free Fall Drop.

c. Pendulum-Impact.

d. Vibration.

e. Cyclic Exposure.

f. Cornerwise Drop.

g. Free Fall Drop.

h. Pendulum-Impact.

i. Vibration.

Unless otherwise specified (see 6.2), the maximum tolerable accelerations (fragility level) will be 30Gs for each engine model. If the 30G factor is exceeded during any of the above tests, this shall constitute rejection of the container design.

3.4.1 Humidity indicator. Unless otherwise specified (see 6.2), a humidity indicator, conforming to MIL-I-26860 shall be incorporated in the container design.

3.4.2 Pressure relief valve. Unless otherwise specified (see 6.2), a 1.0 psi pressure/vacuum valve, conforming to MIL-V-27166, shall be incorporated into the container design. The valve flow rate shall be 2.5 SCFM (Standard Cubic Feet per Minute) when tested in accordance with MIL-V-27166.

3.4.3 Engine record compartment. A compartment shall be provided in the container for the engine records. The compartment shall be weatherproof if not contained in an enclosure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

3.4.4 Opening. The container shall be capable of opening in the field without special devices being required and without damaging the engine. Directions for opening shall be placed on one of the exterior sides of the container.

3.4.5 Reuse. The container shall not be classified as reuseable and there shall be no accountability records. However, the polyurethane container shall be capable of shipping repairable engines to an overhaul activity, as a Level "C" pack.

3.4.6 Exterior surfaces. The exterior surfaces shall be made from any material that will provide suitable marking surfaces and protection in accordance with the applicable Class. In addition, the surfaces shall be true to the extent that containers can be stacked (see 3.4.8).

3.4.7 Materials handling. The container shall be so designed that it may be lifted by forklift truck without permanent deformation to the container or damage to its contents. This may be accomplished by the use of skids. On containers, whose gross weight will exceed 100 pounds, the use of skids is mandatory. Minimum height of skids shall be 2-7/8 inches, and for containers exceeding a gross weight of 350 pounds shall permit four-way lift entry. When specified (see 6.2), features shall be incorporated in the container design to enable the container to be manually handled in areas where handling devices are not available, such as in austere field operations.

3.4.8 Stacking. Containers shall be capable of stacking one upon another in such a manner as not to interfere with forklift access. Each container shall be capable of supporting similar containers (with contents) in a superimposed position (see 4.6.4).

3.5 Workmanship. The container shall be free from any defects which may affect durability, strength, and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with applicable test conditions in section 4.6.

4.4 First article inspection.

4.4.1 Test sample. The test sample shall consist of one container, from each class, representative of a production run. If the containers are foamed-in-place, an additional block having dimensions of 12 by 12 by 12 inches, shall be set aside for the examination of product and density test. Also, if the container is made up of more than one layer of foam, the test block shall have more than one layer to check for adherence between layers.

4.4.2 Test report. The contractor, upon completion of the first article tests, shall prepare a first article test report in accordance with MIL-STD-831 and furnish 3 copies of the report to the procuring activity.

4.4.3 First article tests. For Class 1 containers, the first article tests shall consist of all the tests of this specification. For Class 2 containers, all the tests of this specification are required, except cyclic exposure.

4.5 Quality conformance inspection. Quality conformance inspection shall be in accordance with the provisions of MIL-STD-105, Level II with an acceptable quality level (AQL) of 2.5, except where otherwise indicated. Quality conformance inspection requirements are specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

Table I. Quality Conformance Inspection.

Inspection	Requirement Paragraph	Test Paragraph
Materials	3.2	4.6.1
Uniformity	3.2.1	4.6.1
Thickness	3.2.2	4.6.2
Density	3.2.3	4.6.3
Preservation	3.3.1	4.6.1, 5.1
Design and Construction	Section 3.4	4.6.1, 4.6.4, 4.6.5, 4.6.6, 4.6.7, 4.6.8
Stacking	3.4.8	4.6.9
Uniformity	3.5	4.6.1

4.5.1 End item inspection. A lot shall consist of all containers of the same class offered for inspection at one time.

4.6 Test methods.

4.6.1 Examination of product. The polyurethane container shall be inspected to determine compliance with the requirements of the approved drawing(s), as specified for materials, design, dimensions, identification markings, and all other requirements specified herein for which no tests are specified (see Table II).

Table II. Classification of defects.

Examine	Defect	Category Major	Minor
Materials	Not as specified----------	x	
	Openings in material larger than 1 inch----------	x	
	Openings not seperated by firm material of 1 inch minimum thickness	x	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

Table II. Classification of Defects (cont'd).

Examine	Defect	Category Major	Category Minor
Materials (cont'd)	Accumulations of unexpanded resin		x
Thickness	All surfaces of the engine are not surrounded by foam----------	x	
	Foam in direct contact w/ engine--	x	
Density	Foam density less than 1.3 or greater than 2.5 lbs per sq inch	x	
Design and construction	Preservation is not as specified--	x	
	Humidity indicator not as specified	x	
	Pressure relief valve not as specified----------	x	
	Engine record compartment not provided----------	x	
	Engine record compartment not in an enclosure and not weatherproof---	x	
	Container not stackable----------	x	
	Container not designed to be lifted by a forklift==========	x	
	Skids not present when gross weight of container exceeds 200 lbs-----	x	
	Minimum height of skids less than 2-7/8 inches----------	x	
	Four way lift entry skid not present when gross weight of container exceeds 350 lbs----------	x	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

4.6.2 Thickness. Prefoamed containers shall be inspected any time prior to closing, to insure that the required minimum thickness of foam cushions all surfaces of the engine. On foamed-in-place packs, the pack shall be visually inspected prior to and during the foaming operation to insure that all surfaces of the engine are surrounded by the required minimum thickness of foam.

4.6.3 Density. Density shall be tested in accordance with MIL-P-26514. If the container is foamed-in-place, this test shall be conducted on the rectangular block referenced in 4.4.1. If the container is prefoamed, the foam density may be certified by the foam supplier.

4.6.4 Cornerwise drop test. The loaded container shall be subjected to the cornerwise drop test at temperature extremes, -40 degree F and +140 degree F, in accordance with FED-STD-101, method 5005.1, section 6.1, paragraph 2 (see 3.4).

4.6.5 Free Fall drop test. The container shall be raised and allowed to fall freely from a height of 18 inches to a concrete or similarly hard surface, landing flat on its bottom (see 3.4). Tests shall be conducted at temperature extremes of -40 degree F and +140 degree F (see FED-STD-101, method 5007.1, section 6.3, paragraph 2).

4.6.6 Pendulum impact test. The loaded container shall be subjected to the pendulum impact test of FED-STD-101, method 5012 or incline impact, method 5023 (see 3.4).

4.6.7 Vibration test. The loaded container shall be subjected to the vibration test of MIL-STD-648, section 5.3 (see 3.4).

4.6.8 Cyclic exposure test. The loaded container shall be subjected to the cyclic exposure test in accordance to MIL-P-116, "test B" (see 3.4).

4.6.9 Superimposed-load test. Superimposed-load test shall be performed in accordance with FED-STD-101, method 5016.

5. PACKAGING

5.1 Preservation. The engines in Class 1 containers shall be preserved by Method II of MIL-P-116 in bags fabricated from materials conforming to MIL-B-22191, Type I or equivalent. Engines in Class 2 containers shall be wrapped in a bag fabricated from material conforming to MIL-B-22191, Type II or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The containers covered by this specification are intended for use in the preservation, storage and shipment of small engines not exceeding 500 pounds such as aircraft, helicopters, marine, vehicular and ground support equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class required (see 1.2).

c. Maximum tolerable acceleration in Gs (fragility level) of the engine (see 3.4).

d. Provisions for manual handling when required (see 3.4.6).

e. Type of humidity indicator required (see 3.4.1).

f. Type of relief valve required (see 3.4.2).

g. Specify first article inspection and test methods in the first article test report (see 4.4.2).

h. Specify the number of first article units required (see 6.3).

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of a specified number of units determined during procurement (see 6.2f). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metric conversion. Metric conversions are compiled in Table III, which conforms to ASTM E380.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:07Z
Check the source to verify that this is the current version before use.

TABLE III. Metric Conversions.

English	Metric	Reference
500 lbs.	226.8 kg	1.1
1 in.	2.5 cm	3.2.1
1.3 lbs/cu. ft.	21 kg/cu. m	3.2.3
2.5 lbs/cu. ft.	40 kg/cu. m	3.2.3
100 lbs.	45.4 kg	3.4.7
2-7/8 in.	7.303 cm	3.4.7
350 lbs.	158.8 kg	3.4.7
12 in.	30 cm	4.4.1
18 in.	46 cm	4.6.5
500 lbs.	226.8 kg	6.1

6.5 Subject term (key word) listing.
Container
Container, for small engines
Container, polyurethane, rigid or flexible
Container, rigid, non-reuseable
Container, flexible, non-reuseable
Foam-in-place container (FIP)
Packaging, container
Packaging
Polyurethane

6.6 Changes from the previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

CUSTODIAN

AIR FORCE - 69

REVIEW

AIR FORCE - 71

PREPARING ACTIVITY

AIR FORCE - 69

PROJECT NO. 8115-F484

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

☆ U.S. Government Printing Office: 1982—360-979/3804

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASH DC

POSTAGE WILL BE PAID BY HQ U.S. AIR FORCE
Air Force Packaging Evaluation Agency
HQ AFLC/DSTZT
Wright-Patterson AFB OH 45433-5999

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-38226B	CONTAINERS, POLYURETHANE, RIGID OR FLEXIBLE FOR SMALL ENGINES

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:07Z
Check the source to verify that this is the current version before use.